MODULE 44 · EXAM STRATEGY - ALL SECTIONS

Practice Paper 2

Twenty more new questions across all five sections. Several pair with paper 1: the setting is similar, one constraint changes, and so does the answer.

SECTIONS All five MODULE 44 of 45 QUESTIONS 20

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How to use this paper

- Nine seconds per question. **Pause the video** whenever you need longer.
- Read the **ask** first, then hunt for the **qualifier**.
- Feels like a paper 1 question? Find the **one constraint** that changed.
- Write your letter down. The **score slide** maps each question to its section.

> THE TRAP
>
> Feeling familiar is dangerous. The paired questions punish answering from memory.

> The bills table is partitioned by bill date. Most queries filter on one account_id, out of four million, within a date range.
>
> Those queries still scan whole partitions, and cost far more than expected.

What should the team change?

A Re-partition the whole table by account_id instead of bill date

B Create a separate table for every account

C Cluster the table by account_id, keeping date partitioning

D Add a materialized view that holds every account's bills

◆ PAUSE AND DECIDE

What should the team change?

A Partition by account

Four million values break the partition limit.

B Table per account

Four million tables to manage.

C Cluster by account

Blocks pruned inside each date partition.

D Materialized view

A copy of everything, scanned the same way.

THE SIGNAL

High-cardinality filter inside a date range: cluster, and keep the partitions.

A Dataflow streaming job writes billing events into BigQuery. Finance reports found occasional duplicate rows after worker restarts.

Duplicates are unacceptable, and the team wants the BigQuery connector itself to prevent them.

Which write method should the job use?

A The Storage Write API method, STORAGE_WRITE_API

B The at-least-once method, STORAGE_API_AT_LEAST_ONCE

C Legacy streaming inserts, with retries on every error

D File loads every ten seconds, with WRITE_APPEND

◆ PAUSE AND DECIDE

Which write method should the job use?

A Storage Write API
Exactly-once delivery into BigQuery.

B At least once
Lower latency, but duplicates are possible.

C Legacy inserts
Retries can create duplicates.

D Frequent file loads
Far too frequent, and append repeats on retry.

THE SIGNAL

Duplicates unacceptable from Dataflow: the exactly-once Storage Write API method.

> Larkspur's parent group is building a pricing service that takes writes from customers in Europe, North America and Asia.
>
> It needs relational schemas, strong consistency everywhere, and 99.999% availability.

Which database fits best?

A Cloud SQL, with cross-region read replicas on each continent

B Bigtable, with clusters on three continents

C AlloyDB, with read pools in each region

D Spanner, in a multi-region configuration

◆ PAUSE AND DECIDE

Which database fits best?

A Cloud SQL replicas
One primary for writes, replicas lag behind.

B Bigtable clusters
Eventually consistent, not relational.

C AlloyDB read pools
Read scale, one region for writes.

D Spanner multi-region
Relational, externally consistent, five nines.

THE SIGNAL

Global writes, strong consistency, five nines: Spanner multi-region.

> The customers table holds email, phone and postcode alongside non-sensitive columns. Analysts may query the table, but only the support team may read email and phone.
>
> When analysts touch those columns, they must get an error, not masked values.

What should the team set up?

A Row access policies that hide rows holding an email address

B Policy tags on email and phone, with Fine-Grained Reader for support

C Dynamic data masking on email and phone, with Masked Reader for analysts

D An authorised view without those columns, used by everyone including support

◆ PAUSE AND DECIDE

What should the team set up?

A Row policies

Hides rows, not columns.

B Policy tags

Without the role, those columns return an error.

C Dynamic masking

Masked values, which the brief rules out.

D View for everyone

Support loses access too.

THE SIGNAL

Columns restricted, error rather than mask: policy tags without masking.

> At 10:14, a faulty script deleted thousands of rows from the tariff database on Cloud SQL for PostgreSQL.
>
> Point-in-time recovery is enabled, and the last nightly backup ran at 02:00. The team must recover with the least data loss.

What should the team do?

A Restore last night's 02:00 backup over the instance

B Promote the cross-region read replica in Belgium

C Ask Customer Care to recover the rows from fail-safe

D Clone the instance to a point just before 10:14

◆ PAUSE AND DECIDE

What should the team do?

A Nightly backup

Loses eight hours of writes.

B Promote replica

The replica has already applied the delete.

C Fail-safe

A BigQuery feature, not Cloud SQL.

D Point-in-time clone

The database as it was at 10:13.

THE SIGNAL

Undo a recent mistake in Cloud SQL: point-in-time recovery to a new instance.

A supplier drops daily CSV files into an Amazon S3 bucket.

Larkspur needs them loaded into BigQuery each morning, with the least code and no servers to run.

What should the team use?

A. A Compute Engine VM running a nightly script with the AWS CLI

B. A Dataflow batch job, written in Java, reading from S3

C. A BigQuery Data Transfer Service transfer from Amazon S3

D. Datastream, with the S3 bucket configured as its source

◆ PAUSE AND DECIDE

What should the team use?

A VM and script

A server and a script to maintain.

B Dataflow job

Code to write and maintain.

C Data Transfer Service

Scheduled and managed, with no code.

D Datastream

Replicates databases, not files.

THE SIGNAL

Scheduled loads from another cloud's storage, no code: Data Transfer Service.

A small team in the German business runs a few hundred queries a week, mostly in short bursts on Monday mornings.

Monthly scans total a few terabytes. They want the lowest cost and no capacity to manage.

Which pricing model fits?

A On-demand pricing, paying for the bytes each query scans

B An Enterprise edition reservation, with a one-year commitment

C An Enterprise Plus reservation, with autoscaling

D A Standard edition reservation, with 500 baseline slots

◆ PAUSE AND DECIDE

Which pricing model fits?

A **On-demand**
Pay per query, nothing to manage.

B **Committed Enterprise**
Paying all year for a few busy hours.

C **Enterprise Plus**
Premium features nobody asked for.

D **Standard baseline**
Slots paid for around the clock.

THE SIGNAL

Small, bursty, few terabytes, nothing to manage: on-demand.

> Meter firmware images must be kept for ten years in case of a legal dispute.
>
> They are almost never read, perhaps once in several years. Retrieval fees on that rare read are acceptable.

Which storage class should the team use?

A Coldline, with a lifecycle rule to delete after ten years

B Archive, with a lifecycle rule to delete after ten years

C Nearline, because its retrieval fee is the lowest

D Standard, with Autoclass managing the class later

◆ PAUSE AND DECIDE

Which storage class should the team use?

A Coldline

Fine for quarterly reads, dearer to store.

B Archive

Lowest storage price, still read in milliseconds.

C Nearline

Built for monthly access.

D Standard and Autoclass

Starts at the highest price.

THE SIGNAL

Read less than once a year: Archive. Once a quarter would mean Coldline.

> The meter events topic carries every event type. A new alerts service only needs events whose type attribute is fault, about one percent of traffic.
>
> The team wants the alerts service to receive only those events.

What should the team do?

A Pull every message, and discard non-fault events in the service

B Create a second topic, and republish fault events into it

C Create a subscription with a filter on the type attribute

D Use an ordering key of fault, so those events arrive first

◆ PAUSE AND DECIDE

What should the team do?

A Discard in code

Receives the other ninety nine percent too.

B Republish

A republisher to build and run.

C Subscription filter

Only matching messages are delivered.

D Ordering key

Ordering, not filtering.

THE SIGNAL

Receive only some messages by attribute: a subscription filter.

> Finance wants monthly Google Cloud costs split by team, for BigQuery, Dataflow and Cloud Storage.
>
> Several teams share some projects, so project totals are not enough.

What should the team set up?

A. A separate billing account for each team, moving projects

B. Budget alerts on each project, sent to each team lead

C. A spreadsheet where every team records what it ran

D. Labels on resources and jobs, with billing export

◆ PAUSE AND DECIDE

What should the team set up?

A Billing accounts

Shared projects cannot be split this way.

B Budget alerts

Alerts on spend, not attribution.

C Spreadsheet

Manual, and never complete.

D Labels and export

Costs grouped by team label, in SQL.

THE SIGNAL

Costs by team across shared projects: labels plus billing export.

> Larkspur's marketing team needs churn scores and tariff names from the curated dataset, but must not see personal columns or any other tables.
>
> Both teams are in the same organisation, and nothing should be copied.

How should access be granted?

A An authorised view in their own dataset, with only those columns

B A BigQuery sharing listing that publishes the whole curated dataset

C Data Viewer on the curated dataset for the marketing group

D A nightly scheduled query copying the columns into a marketing table

◆ PAUSE AND DECIDE

How should access be granted?

A Authorised view

Only the chosen columns, no access to the source.

B Sharing listing

The whole dataset, built for other organisations.

C Data Viewer

Every table and column in the dataset.

D Nightly copy

A copy, which the brief rules out.

THE SIGNAL

Another team, a subset, no copies: an authorised view.

> A Bigtable instance stores 400 TB of historical readings, used only by nightly batch analysis jobs.
>
> No user-facing application reads it. The team wants to cut cost, and latency does not matter.

Which storage type fits?

A SSD storage, for its higher read throughput

B HDD storage, for latency-insensitive batch work

C Memorystore in front of Bigtable, to speed up batch reads

D SSD storage with twice as many nodes, for headroom

◆ PAUSE AND DECIDE

Which storage type fits?

A SSD

Pays for speed nobody uses.

B HDD

More terabytes per node, fine for batch.

C Memorystore

An in-memory cache for 400 TB of batch data.

D More SSD nodes

More cost, not less.

THE SIGNAL

Large, batch-only, latency irrelevant: Bigtable HDD storage.

> Auditors require that encryption keys for settlement data in BigQuery and Cloud Storage are controlled by Larkspur, rotated every 90 days, and can be disabled at once in an emergency.
>
> The keys may be held inside Google Cloud.

What should the team use?

A Google default encryption, which already rotates its keys

B Customer-supplied keys, sent with every request

C Cloud EKM, with keys held by an external partner

D CMEK in Cloud KMS, with a 90-day rotation schedule

◆ PAUSE AND DECIDE

What should the team use?

A Default encryption

Google controls it. Larkspur cannot disable it.

B Customer-supplied

Not supported by BigQuery.

C Cloud EKM

Works, but adds a partner nobody asked for.

D CMEK

Larkspur controls, rotates and can disable.

THE SIGNAL

Your control, rotation and disable, inside Google: CMEK.

> The operations team launches the same Dataflow pipeline for different regions and dates.
>
> They do not write code, and want a parameterised job they can start from the console or the command line.

How should engineering package the pipeline?

A As a Flex Template, with region and date as parameters

B As a Python script, with instructions for editing it

C As an Airflow DAG that operations rewrite for each region

D As a container image that operations run on their own laptops

◆ PAUSE AND DECIDE

How should engineering package the pipeline?

A Flex Template
Launched with parameters, no code.

B Editable script
Editing code is what they cannot do.

C Rewritten DAGs
Rewriting DAGs is still code.

D Laptop container
Laptops are not a production platform.

THE SIGNAL

Non-developers launching a pipeline with parameters: a Flex Template.

> On the etl reservation, nightly jobs have started finishing late.
>
> The team wants slot usage per job and per hour across the last two weeks, to find what changed.

Where should they look first?

A Cloud Logging, searching job logs for the word slow

B The billing export, grouped by day

C The jobs timeline views, or the resource charts

D Each query's execution plan, opened one job at a time

◆ PAUSE AND DECIDE

Where should they look first?

A Cloud Logging
Logs, not slot usage over time.

B Billing export
Cost per day, not slots per job.

C Jobs timeline
Slots per job and per period, over weeks.

D Execution plans
One job's detail, not the trend.

THE SIGNAL

Slot use over time, per job: the jobs timeline views and resource charts.

> The customer app checks a user's session and tariff on every page view, adding 40 milliseconds of database time.
>
> The data can be rebuilt from the database if lost. The team wants sub-millisecond reads.

What should the team add?

A. A Cloud SQL read replica in the same region as the app

B. A Memorystore cache in front of the database

C. A BigQuery BI Engine reservation

D. A Firestore database holding a copy of every session

◆ PAUSE AND DECIDE

What should the team add?

A Read replica

Still a database read, in milliseconds.

B Memorystore

In memory, sub-millisecond, and rebuildable.

C BI Engine

For dashboards, not app lookups.

D Firestore copy

A second database, not a cache.

THE SIGNAL

Sub-millisecond reads of rebuildable data: Memorystore.

> Engineers want to type a fault description and find the most similar past repair notes, by meaning rather than keywords.
>
> Two million repair notes are stored in BigQuery.

What should the team build?

A Note embeddings, a vector index, and VECTOR_SEARCH

B A LIKE query with wildcards around every typed keyword

C A k-means model clustering the notes

D Full-text search with search indexes on the notes

◆ PAUSE AND DECIDE

What should the team build?

A Embeddings and vector search

Nearest neighbours by meaning, at speed.

B LIKE query

Exact words only, and slow.

C k-means

Groups notes, but does not rank them for a query.

D Full-text search

Keywords, not meaning.

THE SIGNAL

Similar by meaning: embeddings, a vector index, and vector search.

> The readings pipeline must attach each meter's tariff to every reading. The tariff table has 50,000 rows, and changes about once a month.
>
> Calling the tariff database for every reading has overloaded it.

What should the pipeline do instead?

A Call the database for every reading, with more retries

B Join readings and tariffs with CoGroupByKey in each window

C Load the tariffs as a side input for the enrichment step

D Write readings to BigQuery, then join tariffs in a nightly query

◆ PAUSE AND DECIDE

What should the pipeline do instead?

A More retries

More load on an overloaded database.

B CoGroupByKey

Treats a small, slow table like a stream.

C Side input

Small and slow-changing, held beside the step.

D Nightly join

Enrichment a day late.

THE SIGNAL

Small, slow-changing lookup data: a side input.

Larkspur's billing, metering and customer teams each want to own and publish their own data products.

A central team must still enforce common quality and access rules, and let everyone discover what exists.

Which design fits best?

A One central data team building every table in a single dataset

B Each team in its own project, with no shared catalogue at all

C Copy every team's data into one central lake each night

D A Knowledge Catalog lake per domain, with central policies

◆ PAUSE AND DECIDE

Which design fits best?

A Central team

Central ownership, the opposite of the brief.

B No catalogue

Ownership, but nothing to discover or govern.

C Central copies

Copies, and central ownership again.

D Lake per domain

Domains own data, rules and search are shared.

THE SIGNAL

Domain ownership with central governance: a data mesh in Knowledge Catalog.

> In a Managed Airflow environment, DAGs sometimes vanish from the UI, and the logs show DAG import timeouts.
>
> Several DAG files query BigQuery at the top level, to build their task lists.

What should the team fix first?

A Add more workers and a larger environment size

B Move the BigQuery calls out of top-level code

C Raise the import timeout until the DAGs load

D Split each DAG file into ten smaller files

◆ PAUSE AND DECIDE

What should the team fix first?

A More workers

Workers do not parse DAG files.

B Remove top-level calls

Every parse stops running slow queries.

C Raise the timeout

Hides the cause, and slows every parse.

D Split the files

More files, the same slow calls.

THE SIGNAL

DAG import timeouts: slow top-level code. Move the work into tasks.

Paper 2 answers

QUESTIONS	ANSWERS
1 to 5	C, A, D, B, D
6 to 10	C, A, B, C, D
11 to 15	A, B, D, A, C
16 to 20	B, A, C, D, B

REMEMBER

For every miss, name the phrase you overlooked. That habit is worth more than the score.

Where your marks came from

SECTION	QUESTIONS	REVISIT
1 Design, 22%	3, 7, 13, 19	Modules 02 to 09
2 Ingest and process, 25%	2, 6, 9, 14, 18	Modules 10 to 18
3 Store, 20%	1, 8, 12, 16	Modules 19 to 26
4 Analyse, 15%	4, 11, 17	Modules 27 to 32
5 Maintain and automate, 18%	5, 10, 15, 20	Modules 33 to 39

THE PATTERN

Compare with paper 1. A section weak twice is the one to study next.

Same setting, one constraint changed

PAPER 1	PAPER 2	WHAT CHANGED
Coldline, Q8	Archive, Q8	Read quarterly, or once in years
Row access policy, Q4	Policy tags, Q4	Rows per person, or columns
Sharing listing, Q16	Authorised view, Q11	Another company, or another team
Partition expiration, Q13	Clustering, Q1	Drop by age, or filter by key

THE TIEBREAKER

If you answered a pair the same way twice, reread the constraint that changed.

Module 44 in five lines

- High-cardinality filter in a date range: **cluster**, keep the partitions.
- Duplicates unacceptable from Dataflow: the **Storage Write API** method.
- Columns denied, not masked: **policy tags**. Another team, a subset: **authorised view**.
- Recent Cloud SQL mistake: **point-in-time recovery**. A replica copies the mistake.
- Familiar setting? Find the **one constraint** that changed.

> COMING UP
>
> Next: module 45, practice paper 3, the final twenty, and what your three scores say.